RUNQIU BAO (潤秋 包)

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Visa at Japan: Permanent Resident

Summary

- Computer vision engineer & software developer with 8+ years of experience.
- Proven track record in developing advanced production software for Tier-1 manufacturers across Japan.
- Deep expertise in C++, python, PyTorch-based deep learning.
- Highly reliable team player with focus on aligning technical roadmaps with evolving project goals in dynamic, cross-functional environment.
- Analytical thinker dedicated to solving complex and unstructured problems in physical AI.

Work History

Computer Vision Engineer & Software Developer Oct. 2021 to March 2026
Department of Development, Mujin Inc.

- 3D Object Detector & Computer Vision Software Development:
    - **3D Object Detector**: Developed object detectors for robotic manipulation including
        - A metal piece detector using CAD models as "geometric prompts" to do zero-shot detection.
        - A stereo object detector using specific hand-labeled dataset to train.
        - A general contour-based detector developed by knowledge distillation from foundation models.

      Optimize detection pipelines for real-time performance on both CPU (muti-threading, AVX) and CUDA-enabled systems (CUDA kernel, TensorRT).
    - **Computer Vision Libraries**: Maintaining and optimizing libraries including PCL, OpenCV, Open3D, Genesis engine, and internal libraries.
- Core Systems & Fundamental Works:
    - **Web UI Development for Robotics Platform (Vision Debugging Infrastructure)**: Co-designed and Implemented APIs, schemas, structured log outputs to enable real-time Web UI visualization of diverse intermediate outputs, including images and point cloud data.
    - **Simulation-based Tests**: Initiated and led the integration of a differentiable physics engine and renderer into the internal dev simulator. Created simulation-based CI tests for detector development, covering challenging scenarios such as cluttered metal piles and deformable objects (e.g., thin sheets).
    - **Vision Foundation Models**: Integrated Meta's SAM series models for commercial usage in production; Speeding up SAM-based model inference on CPU-only machines by pruning and distillation, reducing 37% CPU latency.

Image Lab Intern Jan. 2020 to Oct. 2021
HUAWEI Tokyo Research Institute

- Developing dynamic imaging sensors and computational photography algorithms.

HESAI tech. Intern Jul. 2018 to Aug. 2018
Software group, HESAI tech.

- Developing Liar SLAM software.

Education

Master of Engineering Sep. 2018 to Sep. 2020
Precision Engineering, School of Engineering
The University of Tokyo, Tokyo, Japan
Research theme: Global Pose Estimation of Construction Vehicle in GNSS-denied Dynamic Environments Using Stereo Visual SLAM

Exchange Program Oct. 2017 to Apr. 2018
School of Informatics
Technical University of Munich, Munich, Germany

Bachelor of Engineering Sep. 2013 to Jun. 2018
Automotive Engineering (Electronics track), School of Automotive Studies
Tongji University, Shanghai, China
GPA: 4.53/5.00 Rank: 37/196(18.9%)

Honor

Awards:

Title	Organization	Year
Outstanding Master's Thesis Award	Department of Precision Engineering, University of Tokyo	September, 2020
Outstanding Graduate of Tongji University	Tongji University	June, 2018
The Second Class Prize of Mathematical Competition in Modeling (Group Award)	Tongji University	Apr., 2015
The National Second Class Prize of RoboMaster Robot Competition in China (Group Award)	DJI Innovation Company Chinese youth league	Jul., 2015

Skills

Languages: Chinese- Native, English **-** TOEFL 100, Japanese - JLPT N1
Programming Languages: C/C++, Python, React, Matlab, etc.

Publications

Journal Publications

[1] **Runqiu Bao**, Ren Komatsu, Renato Miyagusuku, Masaki Chino, Atsushi Yamashita and Hajime Asama: "Stereo Camera Visual SLAM with Hierarchical Masking and Motion-state Classification at Outdoor Construction Sites Containing Large Dynamic Objects," Advanced Robotics, 2021. (peer-reviewed)
[link]

International Conferences & Symposiums

[1] **Runqiu Bao**, Ren Komatsu, Renato Miyagusuku, Masaki Chino, Atsushi Yamashita and Hajime Asama: "Cost-effective and Robust Visual Based Localization with Consumer-level Cameras at Construction Sites," Proceedings of the 2019 IEEE Global Conference on Consumer Electronics (GCCE 2019), pp. 983–985, 2019. (peer-reviewed)
[link]
[2] Runqiu Bao, "Feature-based Stereo Camera Ego-motion Tracking and its application at Construction Sites"， the 4th Tsinghua University-The University of Tokyo Joint Multidisciplinary Symposium, 2019.

Domestic Conferences

[1] 千野 雅紀, **Runqiu Bao**, 小松 廉, Renato Miyagusuku, 山下 淳, 淺間 一: "土工現場における画像を用いた位置情報取得システムの適用可能性," 土木学会令和 2 年度全国大会第 75 回年次学術講演会講演概要集, VI-1112, pp. 1-2, September 2020.